KAIZEN MATHS

Lesson starter

Sequences and nth Terms

nth Term Formulas

Name _____

S1

5, 7, 9, 11, ...

S2

$T_6 = 49, T_{10} = 77$

S3

$T_n = 6n + 6$, find T_{12}

S4

5, 14, 27, 44, ...

Show your working

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ALG-6E0A-L2 · S

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KAIZEN MATHS

Exit ticket

Sequences and nth Terms

nth Term Formulas

Name _____

E1

6, 13, 22, 33, ___, ___

E2

4, 13, 30, 55, ...

E3

11, 14, 17, 20, ...

KAIZEN MATHS

Exit ticket

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6, 13, 22, 33, ___, ___

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Teacher answers & working

Sequences and nth Terms · nth Term Formulas

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S1

5, 7, 9, 11, ...

Answer:

$2n + 3$

Working:

Find the first differences:

$7 - 5 = 2$

$9 - 7 = 2$

$11 - 9 = 2$

Common difference = 2

Formula: nth term = 2n + c

Using the 1st term: 2(1) + c = 5

$c = 5 - 2 = 3$

nth term = 2n + 3

S2

$T_6 = 49, T_{10} = 77$

Answer:

$7n + 7$

Working:

Given $T_6 = 49$ and $T_{10} = 77$.

For a linear sequence, $T_n = an + b$.

Change in terms: 77 - 49 = 28

Change in position: 10 - 6 = 4

$a = 28 \div 4 = 7$

Substitute: $7 \times 6 + b = 49$

$b = 7$

$T_n = 7n + 7$

Teacher answers & working

Sequences and nth Terms · nth Term Formulas

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S3

$T_n = 6n + 6$, find T_{12}

Answer:

78

Working:

Given $T_n = 6n + 6$.

Substitute n = 12.

$T_{12} = 6(12) + 6$

$T_{12} = 72 + 6$

$T_{12} = 78$

S4

5, 14, 27, 44, ...

Answer:

$2n^2 + 3n + 0$

Working:

Step 1: Find first differences.

First differences: 9, 13, 17

Step 2: Find second differences.

Second differences: 4, 4

Second differences are constant, so the sequence is quadratic.

Second difference = 2a, so 2a = 4.

$a = 2$

Using the first terms gives b = 3 and c = 0.

$T_n = 2n^2 + 3n + 0$

E1

6, 13, 22, 33, ___, ___

Answer:

46, 61

Working:

First differences: 7, 9, 11

Second difference: 2

Continue first differences: 13, 15

5th term: 33 + 13 = 46

6th term: 46 + 15 = 61

Teacher answers & working

Sequences and nth Terms · nth Term Formulas

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E2

4, 13, 30, 55, ...

Answer:

$4n^2 - 3n + 3$

Working:

Terms: 4, 13, 30, 55

First differences: 9, 17, 25

Second differences: 8, 8

Second difference = 2a = 8, so a = 4.

Use T_1, T_2, and T_3 to solve for b and c.

$b = -3$, $c = 3$

$T_n = 4n^2 - 3n + 3$

E3

11, 14, 17, 20, ...

Answer:

$3n + 8$

Working:

Find the first differences:

$14 - 11 = 3$

$17 - 14 = 3$

$20 - 17 = 3$

Common difference = 3

Formula: nth term = 3n + c

Using the 1st term: 3(1) + c = 11

$c = 11 - 3 = 8$

nth term = 3n + 8